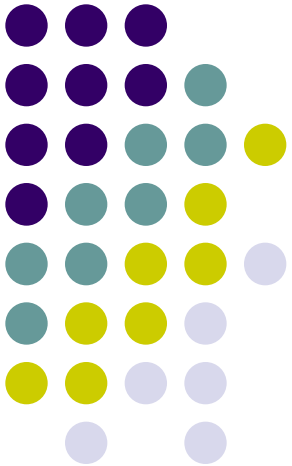
Developing and Validating Scales of Evaluation Use in Multi-site STEM Education Evaluations

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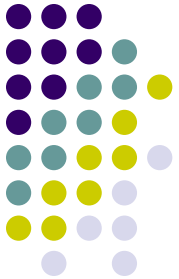


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Research conducted at the University of Minnesota with support from the Beyond Evaluation Use Grant from the National Science Foundation

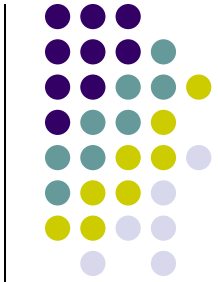
Beyond Evaluation Use:

Determining the Effect of Project Participation on the Influence of NSF Program Evaluations



- NSF grant to study use, influence, and involvement within four NSF-funded programs
- Research team (2 co-PIs and 4 graduate students) based at the University of Minnesota
- Focus on evaluation use and influence in national multi-site STEM program evaluations

Why Validate a Scale for Evaluation Use?



- A valid measure of use and influence will benefit the practice of evaluation by identifying factors critical to evaluation use that will inform evaluation design, and, in turn, affect evaluation practice.
- This study will contribute to research on evaluation use by providing an effective tool for measuring the use and influence of multi-site evaluations.
- Makes a contribution to both evaluation theory and evaluation practice.

Theoretical Framework: Messick's Unitary Concept of Validity

- Validity is "an integrated evaluative judgment of the degree to which empirical evidence and theoretical rationales support the *adequacy* and *appropriateness* of inferences and *actions* based on test scores or other modes of assessment" (Messick, 1989, 1995a, 1995b)
- Compilation of evidence that supports use (or not) of scale in specific setting for specific purpose.
- Not a yes or no question, but stronger or weaker evidence.

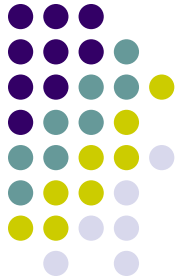
Methods

- Literature review of participatory evaluation research
- Individual key informant interviews with project personnel
- Data collected via online survey

Literature Review

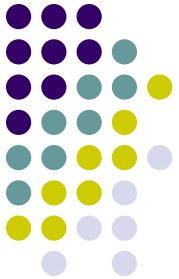
- Fall 2005 – June 2007
- Topic: Participatory evaluation research studies.
- Number of Articles: 27 empirical and theoretical articles directly related to participatory evaluation

In-Depth Interviews



- Topic: Follow-up to survey with questions related to involvement and use/impact
- Wave 1
  - March 2007 – April 2007
  - 17 interviews
- Wave 2
  - May 2008 – August 2008
  - 15 interviews with survey respondents identified as high mean scores for evaluation use

Online Survey



- August 2006 – January 2007
- Topic: Involvement and use in four NSF large scale, multi-site STEM evaluations
- Distributed to 810 PIs & evaluators with valid email addresses
  - An additional 125 names had no valid email address
- Sample Size & Response rate: 46% (372/810).
- Non-respondent survey indicated:
  - No significant differences in levels of involvement between respondents and non-respondents
  - Significant differences in two programs in levels of use/impact

Exploratory Factor Analysis

- Original items submitted to EFA using principal axis factoring with Varimax rotation.
- Eigenvalues greater than 1.0 supported the decision to retain three factors.
- The three factors above 1.0 (15.0, 1.8, and 1.1) explained 78% of the variance.
- Items that loaded at least 0.4 on a factor and were more than 0.2 higher than another factor loading were retained.
- This rule supported the removal of the following two items:
  - Increased skills as a STEM educator.
  - Increased skills as a STEM education evaluator.
- Confirmatory Factor Analysis not conducted due to sample size constraints.

Resulting Factors

Factor 1: Items 25-33 and item 36 focus on using the evaluation to increase knowledge and skills. (10 items)

Factor 2: Items 38-43 are focused on the impact of the evaluation on the respondents' beliefs. (6 items)

Factor 3: Items 45-49 ask about the future use of what respondents learned in the current evaluation. (5 items)

Items 34 and 35 were removed because they did not load disproportionately on one of the factors, but rather had closer to equal factor loadings across all three factors.

The three factors account for nearly 75% of the variance.

- Factor 1 accounts for 29%
- Factor 2 accounts for 25%, and
- Factor 3 accounts for 21%.

Increased knowledge/understanding of:

- planning an evaluation
- implementing an evaluation
- communicating findings
- STEM education
- STEM ed. Evaluation
- my project

Increased skills in:

- planning an evaluation
- implementing an evaluation
- communicating findings
- working on my project

Factor 1: Increased Knowledge, Understanding, and Skills

Increased Belief in Importance of:

- planning
- implementing
- communicating findings
- STEM education
- STEM ed. Evaluation
- my project

Factor 2: Increased Belief in Importance of Evaluation Components

- future evaluation planning
- future use of evaluation plan itself
- future evaluation implementation
- future data collection instruments
- future evaluation communication

Factor 3: Future Use

Items not loading on a factor:

- improved skills as STEM educator.
- improved my skills as a STEM education evaluator.

Messick's Aspect of Validity	Evidence Presented	Strength of Evidence	
		Strong	Weak
Theoretical Soundness	Amount of published research Interview data Data from survey think-aloud testing	X X X	
Internally Consistent	Item variance Scale reliability (alpha=.96)	X X	
Inclusive of Relevant Activities & Processes	Review by experts Scale items mentioned in interviews Types of use (mentioned in literature or interviews) covered by scale	X X	 X
Measures Actual Involvement	Survey Response Distributions Project interviews	X	 X
Statistical Factors Match Rational Structures	Exploratory Factor Analysis Confirmatory Factor Analysis (not viable due to sample size constraints)	X	 X
Differentiates between Groups that are Rationally or Theoretically Different	No significant difference between evaluators and non-evaluator in use		X
Correlates as Expected with Other Variables	Significant, positive correlation with evaluation involvement	X	
Consequential Validity	Discussion of possible biases related to multicultural validity and evaluation use.	X	